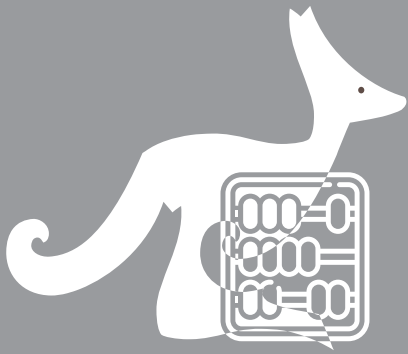




MATH KANGAROO CHINA

考试说明 / Instruction

Time allowed: 75 minutes

考试时长: 75 分钟

Language: English & Chinese

试卷语言: 英文和中文

This is a 30-question multiple-choice competition.

考试包含 30 道单项选择题。

Problems 1 to 10 are worth 3 points each.

1-10 题每题 3 分

Problems 11 to 20 are worth 4 points each.

11-20 题每题 4 分

Problems 21 to 30 are worth 5 points each.

21-30 题每题 5 分

1 point will be deducted for an incorrect answer and
no point for a blank answer.

答错扣 1 分, 不答不扣分。

The full score is 150 (Starting point is 30)

总分为 150 分 (起始分数为 30 分)

Do not open this booklet until you are told to do so.

收到监考老师指令后, 方可打开试卷册。

Mark your answers clearly on the answer sheet using a **2B pencil**.

务必使用 **2B 铅笔** 在答题卡上正确填涂自己的答案。

Calculator cannot be used during the competition.

考试期间禁止使用计算器。

DO NOT discuss problems and solutions in any online or public
forum until April 30.

4 月 30 日前请勿以任何形式公开讨论试题和答案。

**Grade 11
&
Grade 12**

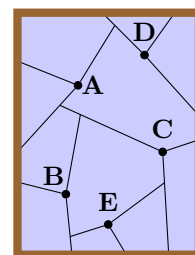
3 points

1. The year 2025 is a perfect square because $2025 = 45^2$. How many years will pass until the next year which is a perfect square?

年份 2025 是一个完全平方数，因为 $2025 = 45^2$ 。请问再过多少年才会出现下一个是完全平方数的年份？

- (A) 25 (B) 91 (C) 121 (D) 500 (E) 2025

2. A student threw five stones, one after the other, which hit a window at the points A , B , C , D and E . Where each stone hits the glass, it creates some linear cracks which stop either at a previous crack or at the boundary. In which order did he throw the stones?



一名学生依次扔出五块石头，分别击中了窗户上的点 A 、 B 、 C 、 D 和 E 。每块石头击中玻璃时，都会产生一些线状裂纹，这些裂纹要么止于先前形成的一条裂纹处，要么止于玻璃的边缘处。请问他是按照什么顺序扔出这些石头的？

- (A) $DACBE$ (B) $ABCDE$ (C) $BDACE$
(D) $BCDAE$ (E) $DCABE$

3. Vasily has 20 different coloured balls either yellow or green or blue or black. Of these, exactly 17 are not green, 15 are not black, and 16 are not blue. How many balls are yellow?

Vasily 有 20 个不同颜色的球，这些球的颜色分别为黄色、绿色、蓝色或黑色中的一种。其中，恰好有 17 个球不是绿色的，15 个球不是黑色的，16 个球不是蓝色的。请问黄色的球有多少个？

- (A) 8 (B) 7 (C) 6 (D) 4 (E) 3

4. In what interval does the value of 88×888 lie?

请问 88×888 的值位于哪一区间内？

- (A) Between 8 and 88 8 和 88 之间
(B) Between 88 and 888 88 和 888 之间
(C) Between 888 and 8888 888 和 8888 之间
(D) Between 8888 and 88888 8888 和 88888 之间
(E) Between 88888 and 888888 88888 和 888888 之间

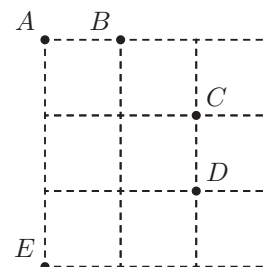
5. Which of the following is equal to the square root of 16^8 ?

请问以下哪个选项等于 16^8 的平方根?

- (A) 4^4 (B) 4^8 (C) 4^{16} (D) 8^4 (E) 8^8

6. Robert wants to choose four points so that the distances between each pair of points are different. Which of the points A , B , C , D and E should be removed?

Robert 想要选出四个点, 使得每两点之间的距离都不相同。请问应该移除点 A 、 B 、 C 、 D 和 E 中的哪一个, 才能满足这一条件?



- (A) A (B) B (C) C (D) D (E) E

7. Mike obtains a number x by dividing the number $\sqrt{41}$ by three. Where is the number x to be found on the number line?

Mike 将 $\sqrt{41}$ 除以 3, 得到一个数 x 。请问这个数 x 在数轴上的哪个位置?

- (A) Between 0 and 1 在 0 和 1 之间
(B) Between 1 and 2 在 1 和 2 之间
(C) Between 2 and 3 在 2 和 3 之间
(D) Between 3 and 4 在 3 和 4 之间
(E) Between 4 and 5 在 4 和 5 之间

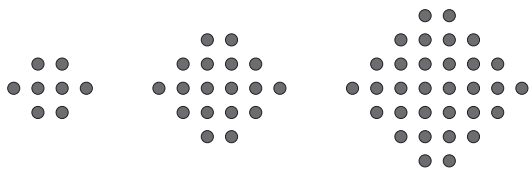
8. Silia's favourite chocolate bars come in packets. Each packet used to contain five bars. Now they contain only four but are sold at the same price. By what percentage has the price of each bar increased?

Silia 最喜欢的巧克力棒是按包出售的。原本每包有 5 根巧克力棒, 但现在每包只有 4 根巧克力棒, 且每包售价保持不变。请问每根巧克力棒的价格上涨了百分之多少?

- (A) by 10% 上涨了 10%
(B) by 20% 上涨了 20%
(C) by 25% 上涨了 25%
(D) by 30% 上涨了 30%
(E) by 50% 上涨了 50%

9. The shapes shown below are the first three shapes of a sequence. How many dots make up the fifth shape in the sequence?

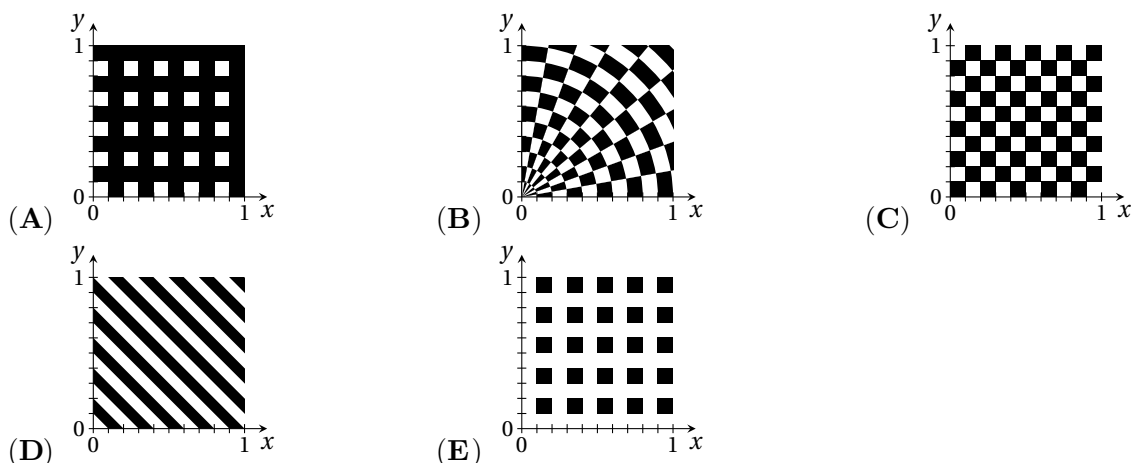
下图展示了一个序列中的前三个形状。请问该序列中的第五个形状由多少个点组成？



- (A) 72 (B) 74 (C) 76 (D) 78 (E) 80

10. In the xy -plane, in the region defined by $0 \leq x \leq 1$, $0 \leq y \leq 1$, some points are painted black. The point (x, y) is painted black if for both x and y the first digit after the decimal point is odd. What does the result look like?

在 xy 平面中，存在一个由 $0 \leq x \leq 1$ 和 $0 \leq y \leq 1$ 定义的区域。在这个区域内，某些点被涂成了黑色。如果 x 和 y 的小数点后第一位数字都是奇数，那么点 (x, y) 就会被涂成黑色。请问这个区域会呈现出怎样的结果？



4 points

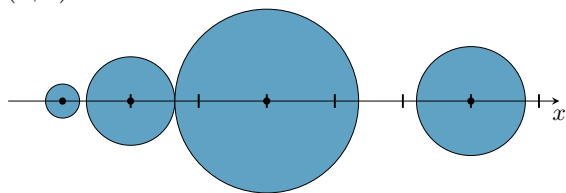
11. Amongst 10 different positive integers, there are exactly five that are divisible by 5 and exactly seven that are divisible by 7. Let M be the largest of these numbers. What is the smallest possible value of M ?

在 10 个不同的正整数中，恰好有五个数能被 5 整除，并且恰好有七个数能被 7 整除。设 M 是这些数中最大的一个数。请问 M 的最小可能值是多少？

- (A) 105 (B) 77 (C) 75
(D) 63 (E) another value 答案为其他值

12. Four circular discs with positive radii r_1, r_2, r_3 and r_4 are centred at $(0, 0)$, $(1, 0)$, $(3, 0)$ and $(6, 0)$. The discs may touch but not overlap. What is the largest possible value of $r_1 + r_2 + r_3 + r_4$?

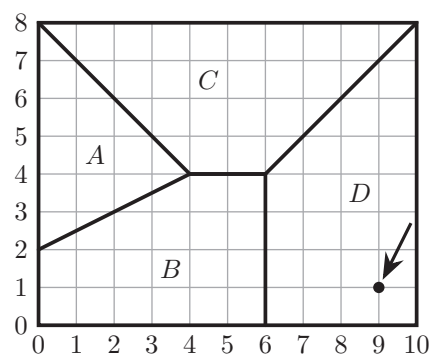
四个半径分别为 r_1, r_2, r_3 和 r_4 的圆盘（半径均为正数），分别以 $(0, 0)$ 、 $(1, 0)$ 、 $(3, 0)$ 和 $(6, 0)$ 为圆心。这些圆盘可以相切但不能相互重叠。请问 $r_1 + r_2 + r_3 + r_4$ 的最大可能值是多少？



- (A) 3 (B) 4 (C) 5 (D) 6
(E) there is no upper limit 没有上限

13. The map shows a small town which has 4 schools. The map shows the regions A, B, C and D of all the points nearest, respectively, to each school. The coordinates of the school in region D are $(9, 1)$. What are the coordinates of the school in region A?

这张地图上展示了一个小镇，该镇共有 4 所学校。地图上划分出四个区域 A、B、C 和 D，每个区域分别包含离某所学校最近的所有点。其中，位于区域 D 的学校的坐标是 $(9, 1)$ 。请问位于区域 A 的学校的坐标是多少？



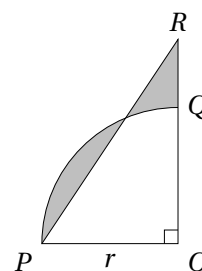
- (A) $(0, 4)$ (B) $(1, 4)$ (C) $(1, 5)$
(D) $(1, 6)$ (E) $(2, 4)$

14. The diagram shows a quarter-circle OPQ and a triangle OPR .

The two shaded regions have the same area. What is the length of OR ?

图中展示了一个四分之一圆 OPQ 和一个三角形 OPR 。

已知两个阴影区域的面积相等，请问 OR 的长度是多少？



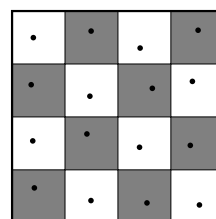
- (A) $\frac{\pi r}{2}$ (B) $\frac{3r}{2}$ (C) πr (D) $\frac{2}{\pi}$ (E) $\frac{\pi}{2r}$

15. The five digit number $\overline{N18NN}$ is divisible by 18. Which of the following statements is true about the digit N ?

五位数 $\overline{N18NN}$ 能被 18 整除。关于数字 N ，请问以下哪个说法是正确的？

- (A) There is exactly one such N 恰好存在一个满足上述条件的 N
- (B) There are exactly two such N 恰好存在两个满足上述条件的 N
- (C) There are exactly three such N 恰好存在三个满足上述条件的 N
- (D) No such N exists 不存在满足上述条件的 N
- (E) There are more than three such N 不止三个 N 可以满足上述条件

16. On a 4×4 giant chessboard there are 16 kangaroos, one in each square. On each turn, each of the kangaroos jumps to a neighbouring square (up, down, left or right, but not diagonally). All kangaroos stay on the board. There can be several kangaroos on any square. After 100 turns, what is the largest possible number of empty squares?



在一个 4×4 的巨型棋盘上，共有 16 只袋鼠，每个格子中各有一只。在每一回合中，每只袋鼠都会跳向相邻的格子（可以向上、向下、向左或向右，但不能对角线跳跃）。所有袋鼠始终留在棋盘上，且每个格子里可以容纳多只袋鼠。请问经过 100 回合后，棋盘上最多可能有多少个空格子？

- (A) 15
- (B) 14
- (C) 12
- (D) 10
- (E) 8

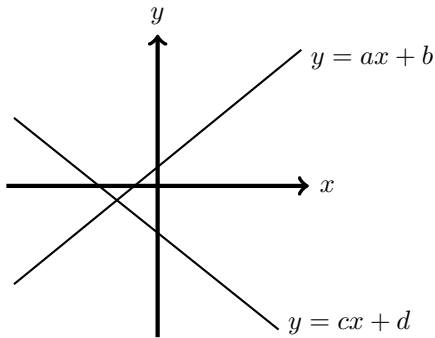
17. What is the smallest positive integer N such that $\sqrt{2\sqrt{3\sqrt{N}}}$ is an integer?

请问满足 $\sqrt{2\sqrt{3\sqrt{N}}}$ 是一个整数的最小正整数 N 是多少？

- (A) $2^{12} \cdot 3^6$
- (B) $2^4 \cdot 3^{14}$
- (C) $2^4 \cdot 3^6 \cdot 5^8$
- (D) $2^4 \cdot 3^2$
- (E) none of the previous 上述答案均不正确

18. A student drew the graphs of two linear functions in a coordinate system, as shown.

一名学生在坐标系中绘制了两个线性函数的图像，如图所示。



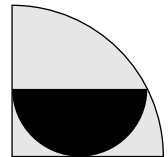
The expression $ac + bd - (ab + cd)$ is always

表达式 $ac + bd - (ab + cd)$ 的值总是

- (A) negative 负数 (B) non-positive 非正数 (C) positive 正数
(D) zero 零 (E) none of the others is always true. 上述选项并非总是成立

19. The area of the black semicircle is 12 cm^2 . What is the area of the big quarter circle?

已知黑色半圆的面积为 12 cm^2 。请问大的四分之一圆的面积是多少？



- (A) 42 cm^2 (B) 36 cm^2 (C) 32 cm^2 (D) 30 cm^2 (E) 25 cm^2

20. When grandma started knitting woollen socks, she had a huge ball of yarn with a diameter of 30 cm. After finishing 70 socks, she still has a ball of yarn with a diameter of 15 cm. How many more socks can grandma knit with the yarn that is left?

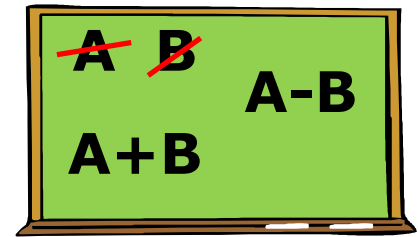
奶奶有一个直径为 30 cm 的大毛线球，她用这些毛线织羊毛袜。织完 70 只袜子后，毛线球的直径变成了 15 cm。请问奶奶还能用剩下的毛线织多少只袜子？



- (A) 70 (B) 50 (C) 30 (D) 20 (E) 10

5 points

21. A student starts with two numbers written on the board. He then deletes them and writes the sum of the numbers and the positive difference of the numbers. He continues the same process with the new numbers. He starts with the numbers 3 and 5 and repeats the process 50 times. What are the two numbers he will end up with?



一名学生在黑板上写下两个数字。然后他擦去这两个数字，并写下这两个数字的和以及它们的正差值。接着，他用新得到的两个数字重复这一过程。如果一开始，这名学生写下的数字为 3 和 5，并重复上述过程 50 次。请问最终他将得到哪两个数字？

- (A) 3^{25} and 5^{25} 3^{25} 和 5^{25}
- (B) 3^{50} and 5^{50} 3^{50} 和 5^{50}
- (C) $2 \cdot 3^{25}$ and $2 \cdot 5^{25}$ $2 \cdot 3^{25}$ 和 $2 \cdot 5^{25}$
- (D) $3 \cdot 2^{25}$ and $5 \cdot 2^{25}$ $3 \cdot 2^{25}$ 和 $5 \cdot 2^{25}$
- (E) none of the previous 上述答案均不正确

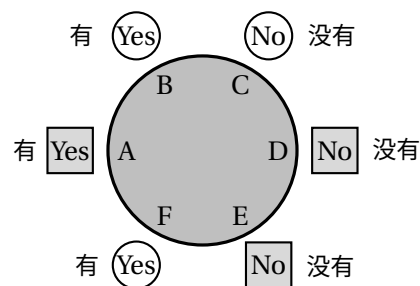
22. Julia and her little sister Paula go out together for a bike ride. Both ride at a constant speed: Julia at 18 km/h and Paula at 12 km/h and they follow the same path. Julia feels tired after 20 minutes and decides to go back. When she meets Paula, Julia tells her to turn around and both return home, each at their own speed. How many minutes later than Julia will Paula arrive home?

Julia 和她的妹妹 Paula 一起外出骑自行车。两人均以匀速骑行：Julia 的速度是 18 km/h，Paula 的速度是 12 km/h，她们沿着同一条路线骑行。骑行了 20 分钟后，Julia 感到疲惫，决定返回。当她遇到 Paula 时，Julia 让 Paula 掉头，然后两人以各自的骑行速度返回家中。请问 Paula 比 Julia 晚多少分钟到家？

- (A) 4
- (B) 6
- (C) 8
- (D) 10
- (E) 15

23. A group of three square men from Mars and a group of three circular men from Jupiter are sitting around a table, as shown. One of the six has the key to their flying saucer. All members of one group always tell the truth and all members of the other group always lie. All six were asked the question “does a person sitting next to you have the key?”

Their answers are shown in the figure. Who has the key?



三位来自火星的方形人和三位来自木星的圆形人围坐在一张桌子旁，如图所示。六人中有一人持有他们的飞碟钥匙。有一个星球的人总是说真话，而另一个星球的人总是说谎话。六个人被问到同一个问题：“坐在你旁边的人有钥匙吗？”他们的回答如图所示。请问谁持有飞碟的钥匙？

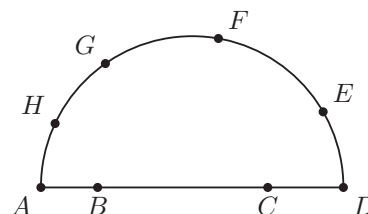
- (A) A (B) B (C) C (D) D (E) E

24. John wrote an arbitrary two-digit integer on a blackboard. Then, he erased the last digit of the integer. As a result, the initial integer decreased by $p\%$. Which of the following is the closest to the largest possible value of p ?

John 在黑板上写下一个任意的两位数整数，然后擦掉了这个数的个位数字。结果，原来的整数减少了 $p\%$ 。请问在以下选项中，哪一个最接近 p 的最大可能值？

- (A) 10 (B) 50 (C) 90 (D) 95 (E) 99

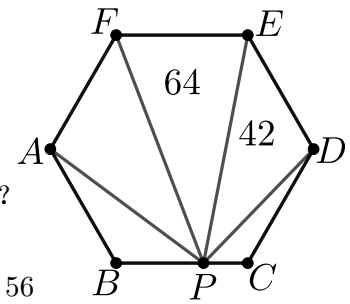
25. On a semicircle with diameter AD , points B and C lie on the diameter and points E, F, G and H lie on the arc of the semicircle. How many triangles can be formed with their vertices at three of these eight points?



在直径为 AD 的半圆中，点 B 和 C 位于直径上，而点 E, F, G 和 H 位于半圆的弧上。请问从这八个点中选三个点作为顶点，可以组成多少个三角形？

- (A) 15 (B) 50 (C) 51 (D) 52 (E) 54

26. The diagram shows a regular hexagon $ABCDEF$. Point P lies on BC so that the area of $\triangle PEF$ is 64 and the area of $\triangle PDE$ is 42. What is the area of $\triangle APF$?



图中展示了一个正六边形 $ABCDEF$ 。点 P 在边 BC 上，使得 $\triangle PEF$ 的面积为 64， $\triangle PDE$ 的面积为 42。请问 $\triangle APF$ 的面积是多少？

(A) 53

(B) 54

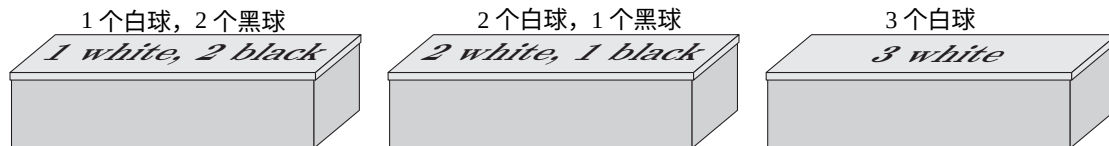
(C) 56

(D) 60

(E) 64

27. Three boxes contain three balls each. The inscriptions on the lids show the contents of each box. The lids are rearranged so that none of them shows the contents correctly.

三个盒子中各装有三个球。盒盖上的标签标明了每个盒子里球的颜色，但盒盖被重新排列，使得没有一个标签与盒内实际装有的球的颜色相符。



Mike picks a box, randomly removes a ball from it, and notes its colour without putting it back. What is the smallest number of balls that Mike needs to remove to determine the contents of each box?

Mike 选择一个盒子，从中随机取出一个球并记录其颜色，取出的球不再放回。请问 Mike 至少需要取出多少个球，才能确定每个盒子里球的颜色？

(A) 0

(B) 1

(C) 2

(D) 3

(E) 4

28. Patricia has written a number in each cell of a 7×10 table. The sum of all the numbers in any 3×4 or 4×3 rectangle is zero. The numbers in two of the cells are shown in the diagram. What is the sum of all the other numbers in the table, excluding the two known numbers?

				20	25				

Patricia 在一个 7×10 表格的每个单元格中都写下一个数字。对于表格中任意一个 3×4 或 4×3 的矩形区域，其中所有数字之和均为零。图中显示了其中两个单元格内的数字。请问除去表格中两个已知数字外，其他所有数字的总和是多少？

(A) -20

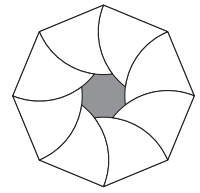
(B) -25

(C) -45

(D) -90

(E) it is not possible to determine 无法确定

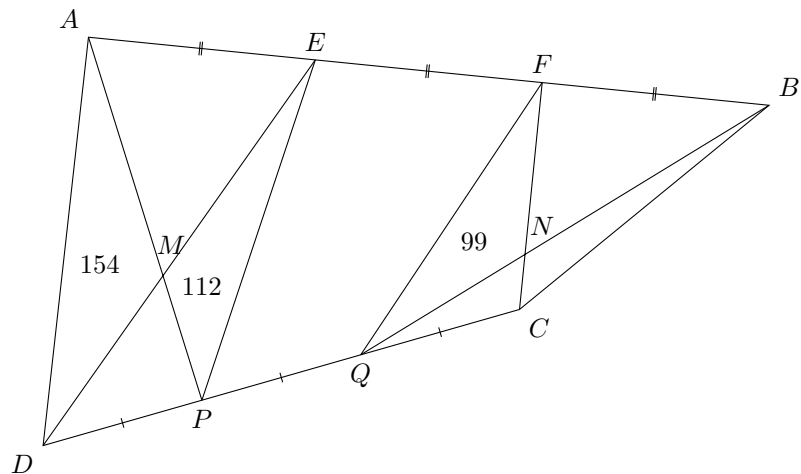
29. The figure shows a regular octagon of side 2 cm. An arc of radius 2 cm has been drawn centred at each vertex, as shown. What is the perimeter of the shaded region?



图中展示了一个边长为 2 cm 的正八边形。以其每个顶点为圆心，分别画一个半径为 2 cm 的弧，如图所示。请问阴影部分的周长是多少？

- (A) π cm (B) $\frac{4\pi}{3}$ cm (C) $\frac{10\pi}{9}$ cm (D) $\frac{6\pi}{5}$ cm (E) $\frac{5\pi}{4}$ cm

30. The sides AB and CD of the convex quadrilateral $ABCD$ are each divided into three parts by points E, F, P and Q so that $AE = EF = FB$ and $DP = PQ = QC$. The diagonals of $AEPD$ and $FBCQ$ intersect at M and N respectively. The areas of triangles AMD , EMP and FNQ are 154, 112 and 99 respectively. What is the area of triangle BCN ?



在凸四边形 $ABCD$ 中，边 AB 和 CD 分别被点 E, F, P 和 Q 分成三等份，使得 $AE = EF = FB$ 且 $DP = PQ = QC$ 。四边形 $AEPD$ 和 $FBCQ$ 的对角线分别相交于点 M 和 N 。已知三角形 AMD 、 EMP 和 FNQ 的面积分别为 154、112 和 99。请问三角形 BCN 的面积是多少？

- (A) 57 (B) 70 (C) 72 (D) 86 (E) 141

Student

3 points

1. The year 2025 is a perfect square because $2025 = 45^2$. How many years will pass until the next year which is a perfect square?

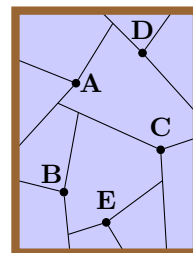
- (A) 25 (B) 91 (C) 121 (D) 500 (E) 2025

SOLUTION: 2025 is the square of 45. So the next square number will be $46^2 = 2116$, which is $2116 - 2025 = 91$ years later.

This is quicker to calculate using the difference of squares: $46^2 - 45^2 = (46 - 45) \cdot (46 + 45) = 1 \cdot 91 = 91$. Another solution is to observe that $46^2 = (45 + 1)^2 = 45^2 + 2 \cdot 45 \cdot 1 + 1^2 = 2025 + 91$.

2. A student threw five stones, one after the other, which hit a window at the points A , B , C , D and E . Where each stone hits the glass, it creates some linear cracks which stop either at a previous crack or at the boundary. In which order did he throw the stones?

- (A) $DACBE$ (B) $ABCDE$ (C) $BDACE$
 (D) $BCDAE$ (E) $DCABE$



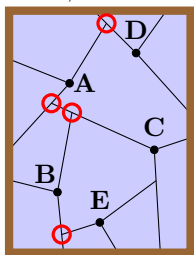
SOLUTION: A crack from E stops at a crack from B , so B was hit before E .

A crack from B stops at a crack from C , so C was hit before B .

A crack from C stops at a crack from A , so A was hit before C .

Finally, a crack from A stops at a crack from D , so D was hit before A .

Thus, the order was $DACBE$.



3. Vasily has 20 different coloured balls either yellow or green or blue or black. Of these, exactly 17 are not green, 15 are not black, and 16 are not blue. How many balls are yellow?

- (A) 8 (B) 7 (C) 6 (D) 4 (E) 3

SOLUTION: There are $20 - 17 = 3$ green balls, $20 - 15 = 5$ black balls, and $20 - 16 = 4$ blue balls, so there are 8 balls that are yellow.

4. In what interval does the value of 88×888 lie?

- (A) Between 8 and 88 (B) Between 88 and 888 (C) Between 888 and 8 888
(D) Between 8 888 and 88 888 (E) Between 88 888 and 888 888

SOLUTION: The value is bigger than $50 \times 200 = 10\,000$, which is bigger than 8 888. The value is smaller than $100 \times 888 = 88\,800$, which is smaller than 88 888, so the value is between 8 888 and 88 888.

5. Which of the following is equal to the square root of 16^8 ?

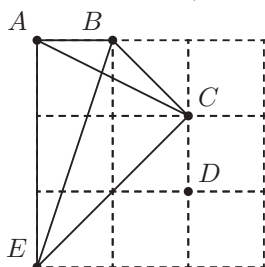
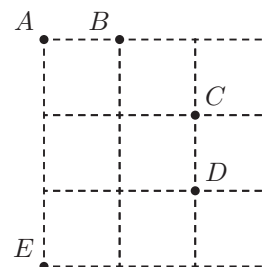
- (A) 4^4 (B) 4^8 (C) 4^{16} (D) 8^4 (E) 8^8

SOLUTION: The square root of 16^8 is equal to $(16^8)^{\frac{1}{2}} = 16^{8 \times \frac{1}{2}} = 16^4 = (4^2)^4 = 4^{2 \times 4} = 4^8$.

6. Robert wants to choose four points so that the distances between each pair of points are different. Which of the points A , B , C , D and E should be removed?

- (A) A (B) B (C) C (D) D (E) E

SOLUTION: When point D is removed, the distances between A , B , C and E are all different, namely 1, $\sqrt{2}$, $\sqrt{5}$, $2\sqrt{2}$, 3 and $\sqrt{10}$.



This is the only option: We can't take B out, since $AC = ED$. We can't take E out either, since $AB = CD$. Since E and B must stay and $ED = DB$, the point D must be removed.

7. Mike obtains a number x by dividing the number $\sqrt{41}$ by three. Where is the number x to be

found on the number line?

- (A) Between 0 and 1 (B) Between 1 and 2 (C) Between 2 and 3
(D) Between 3 and 4 (E) Between 4 and 5

SOLUTION: We first note that $\sqrt{41}$ is between 6 and 7, since $6^2 = 36 < 41 < 49 = 7^2$. Therefore, the number $x = \frac{\sqrt{41}}{3}$ is between $\frac{6}{3} = 2$ and $\frac{7}{3} = 2\frac{1}{3}$, so x is between 2 and 3 on the number line.

8. Silia's favourite chocolate bars come in packets. Each packet used to contain five bars. Now they contain only four but are sold at the same price. By what percentage has the price of each bar increased?

- (A) by 10% (B) by 20% (C) by 25% (D) by 30% (E) by 50%

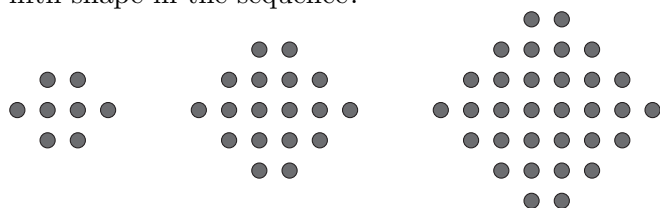
SOLUTION: Let the price of one packet be a . Previously one chocolate bar had the price $\frac{a}{5}$, but now it is $\frac{a}{4}$, which is

$$\frac{\frac{a}{4}}{\frac{a}{5}} = \frac{\frac{1}{4}}{\frac{1}{5}} = \frac{5}{4} = 1.25$$

times the previous price, a 25 % increase.

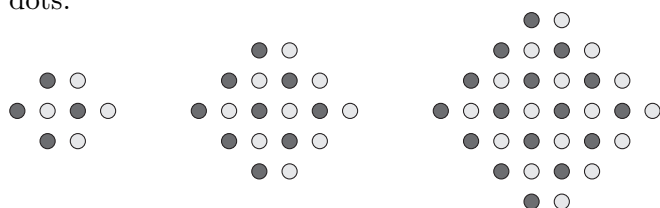
Another approach: Let the price of one packet be \$5. (Since we are discussing percentages, any value will give the same answer). Previously one chocolate bar cost \$1, but now one bar costs $\frac{\$5}{4} = \1.25 , which is a 25 % increase.

9. The shapes shown below are the first three shapes of a sequence. How many dots make up the fifth shape in the sequence?"



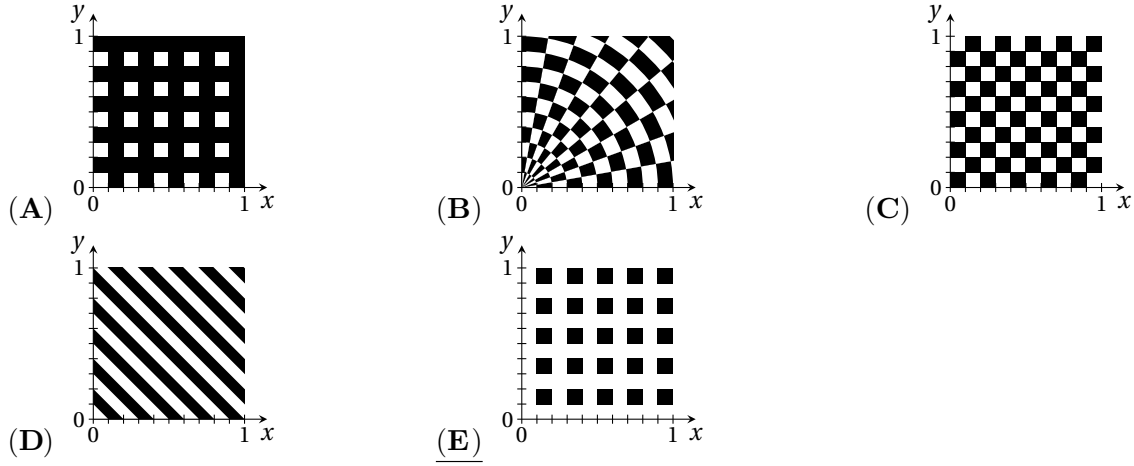
- (A) 72 (B) 74 (C) 76 (D) 78 (E) 80

SOLUTION: Each figure consist of two squares of dots (see below), and the sides of the squares increase by one at each step. The n th figure will have $2(n+1)^2$ dots, so the fifth figure will have $2 \cdot 6^2 = 72$ dots.

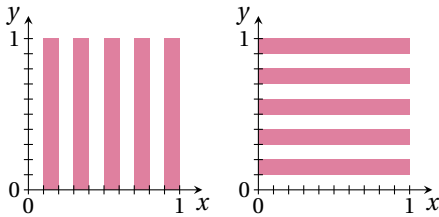


10. In the xy -plane, in the region defined by $0 \leq x \leq 1$, $0 \leq y \leq 1$, some points are painted black. The point (x, y) is painted black if for both x and y the first digit after the decimal point is odd.

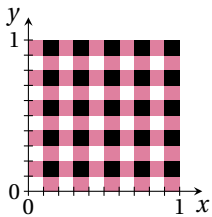
What does the result look like?



SOLUTION: For real numbers between 0 and 1, the first decimal is odd on the intervals $[0.1, 0.2)$, $[0.3, 0.4)$, $\dots$, $[0.9, 1.0)$. This holds for x -coordinates in the colored areas of the following figure on the left, and for y -coordinates on the right.



Both conditions hold in the black areas of the following figure, so the answer E is correct.



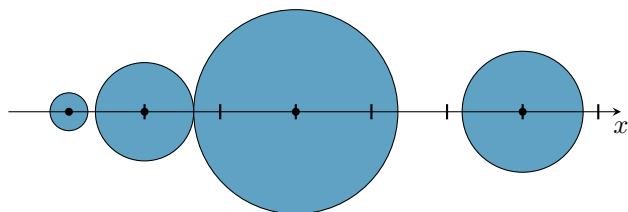
4 points

11. Amongst 10 different positive integers, there are exactly five that are divisible by 5 and exactly seven that are divisible by 7. Let M be the largest of these numbers. What is the smallest possible value of M ?

- (A) 105 (B) 77 (C) 75 (D) 63 (E) another value

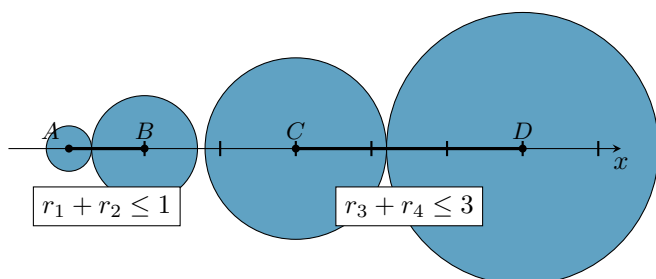
SOLUTION: Since seven of the numbers are divisible by 7 and five of the numbers are divisible by 5, at least two of the numbers are divisible by both 5 and 7. The two smallest such positive integers are 35 and 70, so M is always at least 70. The value $M = 70$ can be achieved for example with the numbers 7, 14, 21, 28, 35, 42, 45, 50, 55, 70, so $M = 70$ is the smallest possible value.

12. Four circular discs with positive radii r_1, r_2, r_3 and r_4 are centred at $(0, 0)$, $(1, 0)$, $(3, 0)$ and $(6, 0)$. The discs may touch but not overlap. What is the largest possible value of $r_1 + r_2 + r_3 + r_4$?



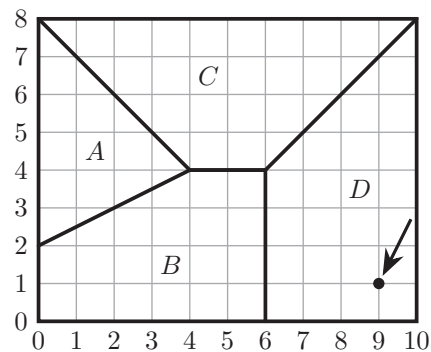
- (A) 3 (B) 4 (C) 5 (D) 6
(E) there is no upper limit

SOLUTION: If we label the center points A, B, C, D from left to right, $r_1 + r_2$ is at most $AB = 1$ and $r_3 + r_4$ is at most $CD = 3$. The maximum sum is thus $1 + 3 = 4$, and it is achieved whenever both circles 1 and 2 and circles 3 and 4 touch. There are multiple ways for this to happen; the circles 2 and 3 need not touch.

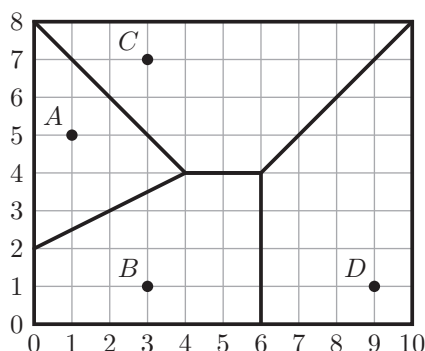


13. The map shows a small town which has 4 schools. The map shows the regions A, B, C and D of all the points nearest, respectively, to each school. The coordinates of the school in region D are (9,1). What are the coordinates of the school in region A?

- (A) (0,4) (B) (1,4) (C) (1,5)
 (D) (1,6) (E) (2,4)



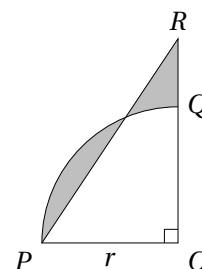
SOLUTION: Each boundary line is a part of the perpendicular bisector of the segment joining two schools. Two neighboring schools are reflections of each other with respect to the boundary. By reflecting the school from region D we find that the school in B is at (3,1). Reflecting this point we find the school in area A to be at (1,5).



14. The diagram shows a quarter-circle OPQ and a triangle OPR . The two shaded regions have the same area. What is the length of OR ?

- (A) $\frac{\pi r}{2}$ (B) $\frac{3r}{2}$ (C) πr (D) $\frac{2}{\pi}$ (E) $\frac{\pi}{2r}$

SOLUTION: Since we are told that the two shaded regions have the same area, the area of the triangle is equal to the area of the quarter-circle. Therefore $\frac{1}{2}r \cdot |OR| = \frac{1}{4}\pi r^2$ and hence $|OR| = \frac{\pi r}{2}$.



15. The five digit number $\overline{N18NN}$ is divisible by 18. Which of the following statements is true about the digit N ?

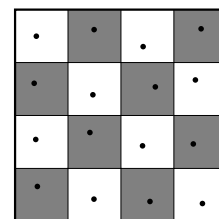
- (A) There is exactly one such N
 (B) There are exactly two such N
 (C) There are exactly three such N
 (D) No such N exists
 (E) There are more than three such N

SOLUTION: A number is even if its last digit is even, so N must be even.

A number is divisible by 9 only if the sum of its digits is divisible by 9. The given number has digital

sum $3N + 9$ so N must be divisible by 3. The only digit which satisfies both criteria is $N = 6$. Hence there is exactly one such N .

16. On a 4×4 giant chessboard there are 16 kangaroos, one in each square. On each turn, each of the kangaroos jumps to a neighbouring square (up, down, left or right, but not diagonally). All kangaroos stay on the board. There can be several kangaroos on any square. After 100 turns, what is the largest possible number of empty squares?



- (A) 15 (B) 14 (C) 12 (D) 10 (E) 8

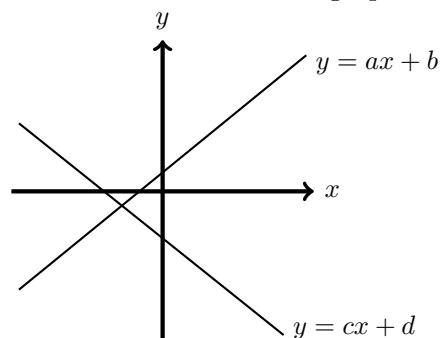
SOLUTION: Each kangaroo changes the color of their square each turn. Since they start on two different colors, there must be at least two squares with kangaroos after each turn, so at most 14 squares can be empty. 14 empty squares is possible: for example all the kangaroos from black squares can go to the bottom left corner and all the kangaroos from white squares to the bottom right corner. This is possible since the number of jumps is even.

17. What is the smallest positive integer N such that $\sqrt{2\sqrt{3\sqrt{N}}}$ is an integer?

- (A) $2^{12} \cdot 3^6$ (B) $2^4 \cdot 3^{14}$ (C) $2^4 \cdot 3^6 \cdot 5^8$ (D) $2^4 \cdot 3^2$ (E) none of the previous

SOLUTION: If $\sqrt{2\sqrt{3\sqrt{N}}} = M$ then $2\sqrt{3\sqrt{N}} = M^2$, so squaring again $2^2 \cdot 3\sqrt{N} = M^4$ and finally $2^4 \cdot 3^2 \cdot N = M^8$. Since the right-hand side is a power of eight, so must be the left-hand side, and by the uniqueness of representation as a product of primes, it follows that the least N is $N = 2^4 \cdot 3^6$.

18. A student drew the graphs of two linear functions in a coordinate system, as shown.



The expression $ac + bd - (ab + cd)$ is always

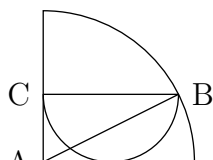
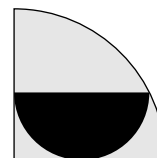
- (A) negative. (B) non-positive. (C) positive.
(D) zero. (E) none of the others is always true.

SOLUTION: The line $y = ax + b$ has positive gradient and positive intercept, so $a > 0$ and $b > 0$. The

line $y = cx + d$ has negative gradient and negative intercept, so $c < 0$ and $d < 0$. Hence we have $ab > 0, cd > 0, ac < 0$ and $bd < 0$. Combining, we see that the given expression is always negative.

19. The area of the black semicircle is 12 cm^2 . What is the area of the big quarter circle?

- (A) 42 cm^2 (B) 36 cm^2 (C) 32 cm^2 (D) 30 cm^2 (E) 25 cm^2



SOLUTION: A

Let r be the length of the radius of the semicircle and R the length of the radius of the quarter circle. Then $|AB| = R$, $|AC| = r$ and $|BC| = 2r$. Pythagoras theorem for triangle ABC gives $r^2 + (2r)^2 = R^2$, so $5r^2 = R^2$. The area of the semicircle is $12 \text{ cm}^2 = \frac{1}{2} \times \pi r^2$, hence $\pi r^2 = 24 \text{ cm}^2$. The area of the quarter circle is $\frac{1}{4} \times \pi R^2 = \frac{1}{4} \times \pi(5r^2) = \frac{5}{4} \times 24 \text{ cm}^2 = 30 \text{ cm}^2$.

20. When grandma started knitting woollen socks, she had a huge ball of yarn with a diameter of 30 cm. After finishing 70 socks, she still has a ball of yarn with a diameter of 15 cm. How many more socks can grandma knit with the yarn that is left?

- (A) 70 (B) 50 (C) 30 (D) 20 (E) 10

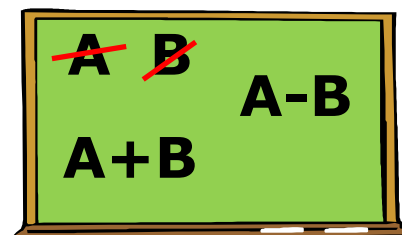


SOLUTION: When the diameter of the yarn ball is halved, only $\left(\frac{1}{2}\right)^3 = \frac{1}{8}$

of its volume is left. Since $\frac{7}{8}$ of the yarn was used for 70 socks, the last $\frac{1}{8}$ is good for 10 more socks.

5 points

21. A student starts with two numbers written on the board. He then deletes them and writes the sum of the numbers and the positive difference of the numbers. He continues the same process with the new numbers. He starts with the numbers 3 and 5 and repeats the process 50 times. What are the two numbers he will end up with?

(A) 3^{25} and 5^{25} (B) 3^{50} and 5^{50} (C) $2 \cdot 3^{25}$ and $2 \cdot 5^{25}$ (D) $3 \cdot 2^{25}$ and $5 \cdot 2^{25}$

(E) none of the previous

SOLUTION: The first two steps are $(A, B) \rightarrow (A + B, A - B)$ and then $\rightarrow ((A + B) + (A - B), (A + B) - (A - B)) = (2A, 2B)$. So every two steps, the numbers double. This leads to answer D.

22. Julia and her little sister Paula go out together for a bike ride. Both ride at a constant speed: Julia at 18 km/h and Paula at 12 km/h and they follow the same path. Julia feels tired after 20 minutes and decides to go back. When she meets Paula, Julia tells her to turn around and both return home, each at their own speed. How many minutes later than Julia will Paula arrive home?

(A) 4

(B) 6

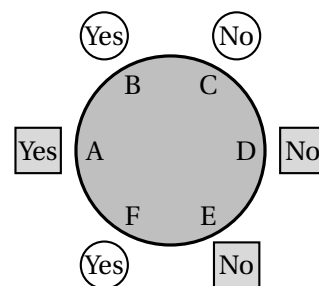
(C) 8

(D) 10

(E) 15

SOLUTION: When Julia turns around, she has travelled 6 km. The total distance that Paula travels is equal to $\frac{12}{18} = \frac{2}{3}$ of that travelled by Julia. If Paula has traveled x km until she meets Julia on her way back, then $x = \frac{2}{3}(6 + 6 - x)$. Solving gives $x = 24/5$. So in total Paula covers back and forth $\frac{48}{5}$ km, and it takes her 48 minutes to do them. Since Julia's travel time is 40 minutes, Paula arrives 8 minutes later.

23. A group of three square men from Mars and a group of three circular men from Jupiter are sitting around a table, as shown. One of the six has the key to their flying saucer. All members of one group always tell the truth and all members of the other group always lie. All six were asked the question "does a person sitting next to you have the key?" Their answers are shown in the figure. Who has the key?



- (A) A (B) B (C) C (D) D (E) E

SOLUTION: If C is telling the truth (so all circular men tell the truth and all square men lie) then neither B or D have the key. As A is lying it means that F does not have the key. As F is saying the truth it follows that E does not have the key, and similarly the (false) information given by E means that D has the key. This is a contradiction because we know that D does not have the key. We conclude that C lies. So the key is held by one of B or D. Given that C lies it means that E tells the truth, so D does not have the key. To conclude, the key is held by B. One can check this, together with the conclusion that circular men lie, is consistent with the problem.

Alternatively:

If circular men are telling the truth, A must have the key, since B and F say they are sitting next to it. However, then the square men E and D would also be telling the truth, which is a contradiction. So the circular men must lie, and the square men must tell the truth. This means that A and C are sitting next to the key, so B has it.

24. John wrote an arbitrary two-digit integer on a blackboard. Then, he erased the last digit of the integer. As a result, the initial integer decreased by $p\%$. Which of the following is the closest to the largest possible value of p ?

- (A) 10 (B) 50 (C) 90 (D) 95 (E) 99

SOLUTION: The best value for p is achieved with the number 19, which decreases $\frac{19-1}{19} \approx 95\%$. Let's prove this rigorously.

Let $n = \overline{ab} = 10a + b$ be the initial two-digit number, where $a \in \{1, 2, \dots, 9\}$ and $b \in \{0, 1, 2, \dots, 9\}$. Then the new number is $m = a$. The initial number decreased by $p\%$, where

$$p = \frac{n - m}{n} \cdot 100 = \frac{9a + b}{10a + b} \cdot 100 = \left(1 - \frac{a}{10a + b}\right) \cdot 100 = \left(1 - \frac{1}{10 + b/a}\right) \cdot 100.$$

However, $10 \leq 10 + b/a \leq 10 + 9/1 = 19$.

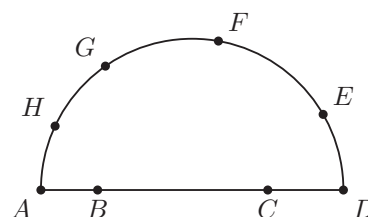
Thus,

$$p = \left(1 - \frac{1}{10 + b/a}\right) \cdot 100 \leq \left(1 - \frac{1}{19}\right) \cdot 100 = \frac{1800}{19} = 94,7... \approx 95.$$

This result is achievable for the number $n = 19$ with $a = 1$ and $b = 9$.

25. On a semicircle with diameter AD , points B and C lie on the diameter and points E, F, G and H lie on the arc of the semicircle. How many triangles can be formed with their vertices at three of these eight points?

- (A) 15 (B) 50 (C) 51 (D) 52 (E) 54



SOLUTION: We have 8 points in the figure: A, B, C, D, E, F, G, H .

A triangle is formed by 3 points. To find the number of possible triangles, we use combinations to select 3 points out of the 8 available points:

$$\binom{8}{3} = \frac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$$

The four points A, B, C, D lie on a straight line, so we need to exclude the triangles formed by these points as they do not constitute real triangles. The number of triangles that can be formed by these points is:

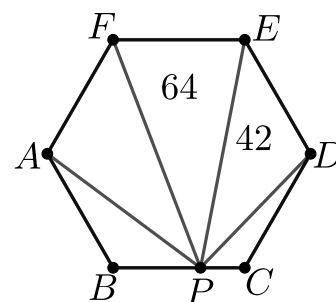
$$\binom{4}{3} = 4$$

So the total number of possible triangles is:

$$56 - 4 = 52$$

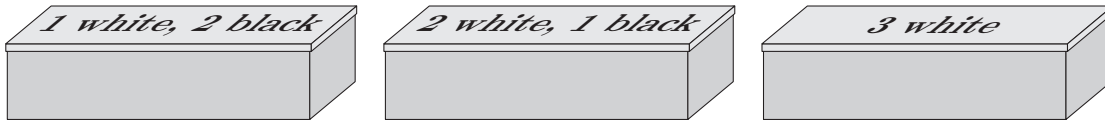
26. The diagram shows a regular hexagon $ABCDEF$. Point P lies on BC so that the area of $\triangle PEF$ is 64 and the area of $\triangle PDE$ is 42. What is the area of $\triangle APF$?

- (A) 53 (B) 54
(C) 56 (D) 60
(E) 64



SOLUTION: The area of $\triangle PEF$ is equal to the area of $\triangle CEF$ (same base and height) and hence is $\frac{1}{3}$ of the area of the whole hexagon. Hence the hexagon has area $3 \times 64 = 192$. The area of $\triangle ADP$ is equal to the area of $\triangle ADC$ which is also $\frac{1}{3}$ of the hexagon. Hence the area of pentagon $AFEDP$ is $\frac{1}{3} + \frac{1}{2} = \frac{5}{6}$ of the hexagon which is 160. Hence the area of $\triangle APF$ is $160 - 64 - 42 = 54$.

27. Three boxes contain three balls each. The inscriptions on the lids show the contents of each box. The lids are rearranged so that none of them shows the contents correctly.



Mike picks a box, randomly removes a ball from it, and notes its colour without putting it back. What is the smallest number of balls that Mike needs to remove to determine the contents of each box?

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

SOLUTION: Box 2 contains either 1 white and 2 black balls or 3 white balls. It is enough to take two balls out of box 2. If both of them are white, then this box must be the one with 3 white balls and the content of the other two boxes is then determined. If either (or both) of the balls are black, then it follows that box 2 must be the one with 1 white and 2 black balls. Again, the content of the other two boxes is now determined.

28. Patricia has written a number in each cell of a 7×10 table. The sum of all the numbers in any 3×4 or 4×3 rectangle is zero. The numbers in two of the cells are shown in the diagram. What is the sum of all the other numbers in the table, excluding the two known numbers?

- (A) -20 (B) -25 (C) -45
 (D) -90 (E) it is not possible to determine

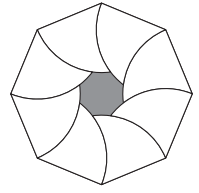
				20	25				

SOLUTION: The table can be divided into five grey rectangles, each with a sum of 0 and a striped area. Since the sum of the numbers in any 3×4 rectangle has to be 0, the sum of the numbers 20 and 25 and the sum of the number in the striped area is also 0. Therefore, the sum of the numbers in the striped area is -45 . The sum of the numbers in the table is $5 \cdot 0 + (-45) = -45$, the sum of all the other numbers in the table, excluding the two known numbers is $-45 - 20 - 25 = -90$.

To demonstrate that such a construction is possible, one example is given below, using a diagonal pattern:

25	-20	-25	20	25	-20	-25	20	25	-20
-20	-25	20	25	-20	-25	20	25	-20	-25
-25	20	25	-20	-25	20	25	-20	-25	20
20	25	-20	-25	20	25	-20	-25	20	25
25	-20	-25	20	25	-20	-25	20	25	-20
-20	-25	20	25	-20	-25	20	25	-20	-25
-25	20	25	-20	-25	20	25	-20	-25	20

29. The figure shows a regular octagon of side 2 cm. An arc of radius 2 cm has been drawn centred at each vertex, as shown. What is the perimeter of the shaded region?

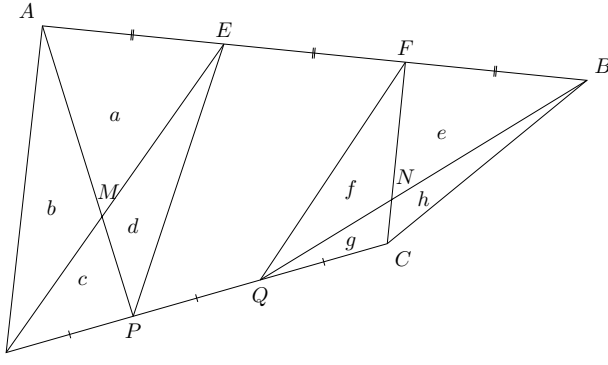
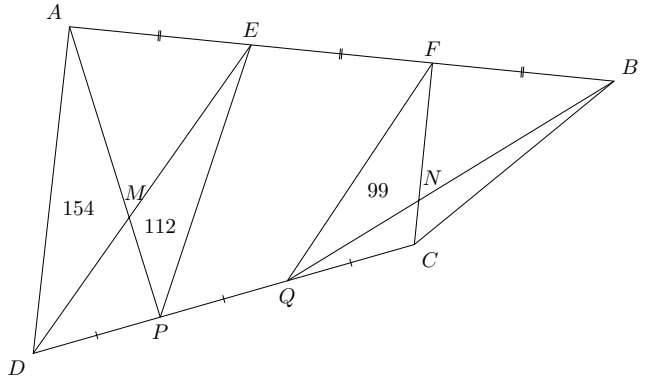


- (A) π cm
- (B) $\frac{4\pi}{3}$ cm
- (C) $\frac{10\pi}{9}$ cm
- (D) $\frac{6\pi}{5}$ cm
- (E) $\frac{5\pi}{4}$ cm

SOLUTION: Each interior angle of the regular octagon is 135° so each arc measures $135^\circ - 60^\circ = 75^\circ$, because each vertex of the central region forms equilateral triangles with the sides of the regular octagon. Thus, the central region consists of 8 arcs, where each one measures $75^\circ - 60^\circ = 15^\circ$. So the perimeter is $\frac{8 \times 15}{360} \cdot 4\pi = \frac{4\pi}{3}$ cm.

30. The sides AB and CD of the convex quadrilateral $ABCD$ are each divided into three parts by points E, F, P and Q so that $AE = EF = FB$ and $DP = PQ = QC$. The diagonals of $AEPD$ and $FBCQ$ intersect at M and N respectively. The areas of triangles AMD , EMP and FNQ are 154, 112 and 99 respectively. What is the area of triangle BCN ?

- (A) 57 (B) 70 (C) 72 (D) 86 (E) 141



SOLUTION: D

Using the labels in the diagram to represent areas of each region, we have:

$$a + b = [ADE] = \frac{[ADB]}{3} \text{ and } g + h = [BCQ] = \frac{[BCD]}{3}$$

$$\text{Combining, we see that } a + b + g + h = \frac{[ABCD]}{3}.$$

Also, we have:

$$a + d = [AEP] = \frac{[AFP]}{2} \text{ and } f + g = [CFQ] = \frac{[CFP]}{2}$$

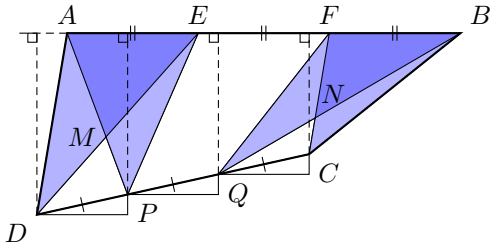
$$\text{Again, combining, we have } a + d + f + g = \frac{[AFCP]}{2} = \frac{[FAC] + [CPA]}{2} = \frac{\frac{2}{3}([BAC] + [CDA])}{2} = \frac{[ABCD]}{3}$$

$$\text{So } a + b + g + h = a + d + f + g \text{ and hence } b + h = d + f$$

$$\text{Using the numbers given, } h = d + f - b = 112 + 99 - 154 = 57.$$

$$[NCB] = 57$$

Alternative solution:



The triangles ADE , APE , FQB , and FCB all have bases of equal lengths (AE and FB) and heights that form a decreasing arithmetic progression. Hence the areas of these four triangles will also decrease in arithmetic progression. In particular $[ADE] - [APE] = [FQB] - [FCB]$. Subtracting the common triangles AME and FNB from each pair of these triangles gives $[AMD] - [EMP] = [FNQ] - [BCN]$. Since $[AMD] - [EMP] = 154 - 112 = 42$, also $99 - [BCN] = 42$, and we get $[BCN] = 99 - 42 = 57$.